FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: resistance
Model status: IEC60287_1_1_2023_CONDUCTOR_AC_RESISTANCE_COMPONENT
Captured UTC: 2026-10-26T06:04:45.777Z
Warning: ENGINEERING SCREENING ONLY ? independently verify actual manufacturer, utility, installation, e
quipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certif
ication or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: r20OhmPerM | Entered value: 0.0001
Input path: alpha20PerC | Entered value: 0.004
Input path: operatingConductorC | Entered value: 90
Input path: frequencyHz | Entered value: 60
Input path: skinFactorKS | Entered value: 0.8
Input path: proximityFactorKP | Entered value: 0.8
Input path: conductorDiameterMM | Entered value: 25
Input path: axisSpacingMM | Entered value: 40
Input path: geometry | Entered value: three-core

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: IEC60287_1_1_2023_CONDUCTOR_AC_RESISTANCE_COMPONENT
Calculated output path: scope | Value: AC resistance only. k_s and k_p and conductor arrangements must b
e verified from cable construction.
Calculated output path: dcAtOperatingTempOhmPerM | Value: 0.00012800000000000002
Calculated output path: skinFactor | Value: 0.004609317510365762
Calculated output path: proximityFactor | Value: 0.007956273855403486
Calculated output path: skinParameterXs | Value: 0.9708129562778495
Calculated output path: acResistanceOhmPerM | Value: 0.00012960839569481851
Calculated output path: acToDcRatio | Value: 1.0125655913657694
Calculated output path: extraACLossPercent | Value: 1.2565591365769246
Calculated output path: geometry | Value: three-core
Calculated output path: notIncluded[0] | Value: Undocumented k_s/k_p / strand type interpolation
Calculated output path: notIncluded[1] | Value: Sheath and armour circulating losses
Calculated output path: notIncluded[2] | Value: Installation thermal resistances and grouping
Calculated output path: notIncluded[3] | Value: IEC 60287 complete ampacity approval

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY ? independently verify actual manufac
turer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: resistance
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:45.777Z
Evidence / limitation: outputs.scope | Value: AC resistance only. k_s and k_p and conductor arrangements
 must be verified from cable construction.